TRAINING AND SCALING LANGUAGE MODELS: FROM FIRST PRINCIPLES TO EFFICIENT SERVING

Session 13: Beyond Dense Transformers

Sparsity, State Space Models, and Architectural Synthesis

BRAIN team

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Lecture Plan

1. Architectural Limits of Dense Attention

PART 1

Architectural Limits of Dense Attention

1 Architectural Limits of Dense Attention

The Scaling Frontier: Sparsity & Alternative Attention

- **Dense Transformer constraints:** Every parameter is activated for every token ($\mathcal{O}(N)$ compute cost), and KV-cache footprint grows linearly with context length T.
- **Sparse Mixture-of-Experts (MoE):** Route each token to Top-k of E specialized MLP experts. Dramatically expands parameter capacity without increasing active FLOPs.
- **Recurrence & State Space Models:** Mamba / RWKV replace softmax attention with linear recurrent states, achieving $\mathcal{O}(1)$ inference memory per token.

